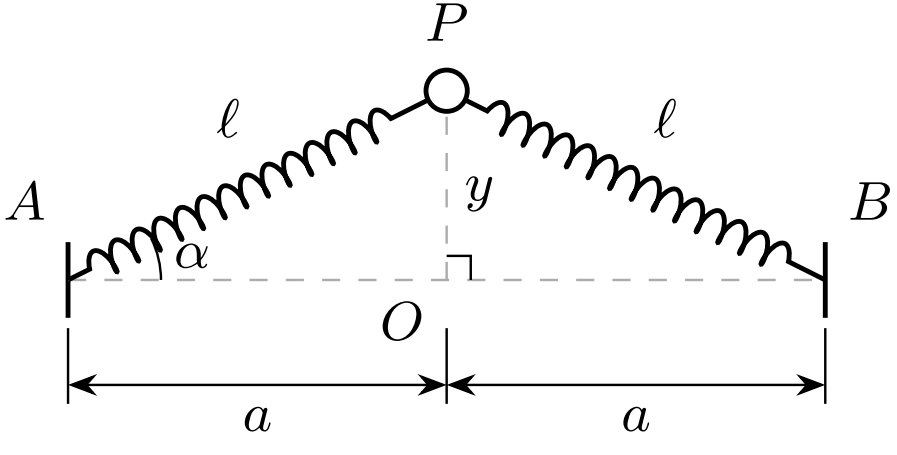
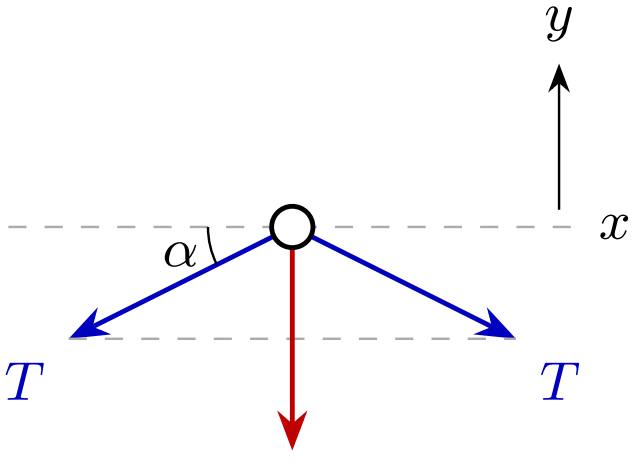
桌面内几何；$\ell^2 = a^2 + y^2$



小球自由体：$y > 0$



$$F_y = -2T \sin \alpha$$